MIL-P-45453B(MU)
7 AUGUST 1963
SUPERSEDING
MIL-P-45453A(Ord)
16 June 1960

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M79 METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of primer designated as Primer, Percussion, M79.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition & Special Weapons, General Specification for.

MIL-P-10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8838012 — Cup, Battery.

8838015 — Plug, Firing.

8838157 — Plug, Closing.

8839479 — Plug.

8839480 — Body, Alternative.

8839482 — Head.

8840616 — Body.

PUBLICATIONS

ORDNANCE CORPS

IEL-8816612 — Index of Inspection Equipment Lists.

FSC 1390

701044-63-(522)

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts. The metal parts shall comply with all requirements specified on Drawings (dwgs.) 8838012, 8838015, 8838157, 8839479, 8839480, 8839482 and 8840616, as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test. The tubing to be fabricated into primer bodies shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing dimensions when tested as specified in 4.4.1 (see 6.4).

3.4 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specification referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during

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the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 each of every individual component and 25 primer body tubes prior to drilling and other machining, which have been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for material. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

4.3.3 *Rejection.* If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 *Lot formation.* A lot shall consist of inert components produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot number.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for

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critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 *Cup, battery.* **(See dwg. 8838012.)**

Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Outside diameter, minimum (min.)	Gage	01001
102.	Thickness through bottom, maximum (max.)	Gage	01002
103.	Length	Gage	01003
Minor:	AQL 2.50 percent		
201.	Diameter of hole	Gage	01004
202.	Thickness through bottom, min	Gage	01005
203.	Inside diameter	Gage	01006
204.	Outside diameter, max	Gage	01007
205.	Evidence of poor workmanship (see 3.4)	Visual	01008

4.3.2.2 *Plug, firing.* **(See dwg. 8838015.)**

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Large diameter, min	Gage	02001
102.	Length and profile of point	Gage	02002
103.	Length of pilot, min	Gage	02003
104.	Total length	Gage	02004
Minor:	AQL 1.50 percent		
201.	Large diameter, max	Gage	02005
202.	Length of pilot, max	Gage	02006
203.	Diameter of pilot	Gage	02007
204.	Evidence of poor workmanship (see 3.4)	Visual	02008

4.3.2.3 *Head.* **(See dwg. 8839482.)**

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 1.50 percent		
101.	Pitch diameter of body threads, min	Gage	03001
102.	Major diameter of body threads, min	Gage	03002
103.	Pitch diameter of case threads, min	Gage	03003
104.	Major diameter of case threads, min	Gage	03004
105.	Diameter of percussion element cavity	Gage	03005
106.	Concentricity of percussion element cavity with pitch diameter of case threads	Gage	03006
107.	Depth to bottom of percussion element cavity	Gage	03007
108.	Diameter of battery cup cavity, max	Gage	03008
109.	Depth of battery cup cavity, min	Gage	03009
110.	Depth of plug cavity, min	Gage	03010
111.	Diameter of plug cavity, min	Gage	03011

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112.	Diameter of body thread undercut, min	Gage	03012
113.	Crack or split	Visual	03013
114.	Flash hole missing or obstructed	Visual	03014
Minor:	AQL 6.50 percent		
201.	Pitch diameter of body threads, max	Gage	03015
202.	Major diameter of body threads, max	Gage	03016
203.	Pitch diameter of case threads, max	Gage	03017
204.	Major diameter of case threads, max	Gage	03018
205.	Thickness of flange	Gage	03019
206.	Diameter of flange, max	Gage	03020
207.	Diameter of flash hole, min	Gage	03021
208.	Diameter of battery cup cavity, min	Gage	03022
209.	Length of body thread undercut, max	Gage	03023
210.	Length of case thread undercut, max	Gage	03024
211.	Concentricity of flash hole with percussion element cavity	Gage	03025
212.	Concentricity of flange with pitch diameter of case threads	Gage	03026
213.	Distance between wrench slots	Gage	03027
214.	Depth of wrench slots	Gage	03028
215.	Length from rear end to bottom of flange	Gage	03029
216.	Length from rear end to forward end of case thread	Gage	03030
217.	Diameter of crimping lip	Gage	03031
218.	Length of crimping lip	Gage	03032
219.	Depth of plug cavity, max	Gage	03033
220.	Diameter of plug cavity, max	Gage	03034
221.	Finish improper	Visual	03035
222.	Plating damaged exposing base metal	Visual	03036
223.	Evidence of poor workmanship (see 3.4)	Visual	03037

4.3.2.4 ***Body and body alternate.*** **(See dwg's 8840616 and 8839480.)**

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Flash hole missing	Gage	04001
2.	Crack or split	Visual	04002
Major:	AQL 0.65 percent		
101.	Pitch diameter of head thread	Gage	04003
102.	Minor diameter of head thread, max	Gage	04004
103.	Thickness through bottom, min. (Alternate only)	Gage	04005
104.	Location of first two pairs of holes	Gage	04006
105.	Diameter of holes	Gage	04007
106.	Pitch diameter of closing plug thread	Gage	04008
107.	Minor diameter of closing plug thread, max	Gage	04009
Minor:	AQL 4.00 percent		
201.	Wall thickness, min	Gage	04010
202.	Length, min.	Gage	04011
203.	Length of head thread, min	Gage	04012
204.	Inside diameter, min	Gage	04013
205.	Outside diameter, max	Gage	04014
206.	Depth of thread counterbore, max	Gage	04015
207.	Diameter of thread counterbore	Gage	04016
208.	Length of closing plug thread, min	Gage	04017
209.	Plating damaged exposing base metal	Visual	04018
210.	Evidence of poor workmanship (see 3.4)	Visual	04019

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4.3.2.5 *Plug, closing.* (See dwg. 8838157.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Pitch diameter of thread, min	Gage	05001
Minor:	AQL 2.50 percent		
201.	Pitch diameter of thread, max	Gage	05002
202.	Major diameter of thread, min	Gage	05003
203.	Total length	Gage	05004
204.	Slot missing	Visual	05005
205.	Evidence of poor workmanship (see 3.4)	Visual	05006

4.3.2.6 *Plug.* (See dwg. 8839479.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Outside diameter, max	Gage	06001
102.	Total length, max	Gage	06002
103.	Flash hole missing or obstructed	Visual	06003
Minor:	AQL 1.50 percent		
201.	Depth of charge cavity	Gage	06004
202.	Diameter of flash hole, min	Gage	06005
203.	Evidence of poor workmanship (see 3.4)	Visual	06006

4.3.3 *Testing.*

4.3.3.1 *Hydrostatic test of primer body (see 3.3) critical defect—Code No. 04020.* This test shall be performed as specified in 4.4.1, using test equipment specified in 4.3.4. Any tubing which fails to meet the specified requirement shall be classed defective and removed from the lot.

4.3.4 *Inspection equipment.* Index of Inspection Equipment Number IEL 8816612 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection equipment.* The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.4 Test methods and procedures

4.4.1 *Hydrostatic test.* Prior to machining, the tubing for primer fabrication shall be

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assembled into the test equipment specified in 4.3.4 and shall be tested for not less than five seconds at not less than the hydrostatic pressure calculated from the following formula:

$$P = \frac{2\,St}{D} \times .9$$

where:

P = test pressure in pounds per square inch (psi)

S = min. specified yield strength of tubing in psi.

t = specified min. wall thickness in inches

D = specified max. diameter in inches

Observation shall be made for rupture, indications of flaws as evidenced by leakage and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 *Level C.* Packing and marking shall be in accordance with Specification MIL–P–10025.

5.2 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the inspection equipment list by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM–608–12.

CDCF — Contractor's design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an inspection equipment list references mandatory

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Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 **Hydrostatic test.** If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.5 **Submission of initial production samples.** Instructions, as to the location for the evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–NI.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings,

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specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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